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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2024

EE2005-1 – NETWORK ANALYSIS

(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Constant voltage source is**
A) Active and bilateral
B) Passive and bilateral
C) Active and unilateral
D) Passive and unilateral
2. **The circuit has resistors, capacitors and semiconductor diodes. The circuit will be known as**
A) Non-linear circuit
B) Linear circuit
C) Bilateral circuit
D) None of the above
3. **Ideal voltage source has**
A) Zero internal resistance
B) Infinite internal resistance
C) Low value of current
D) Large value of e.m.f.
4. **Two inductors of values L_1 and L_2 are coupled by a mutual inductance M . By interconnection of the two elements, one can have a minimum inductance of**
A) $L_1 + L_2 - M$
B) $L_1 + L_2$
C) $L_1 + L_2 - 2M$
D) $L_1 + L_2 + M$
5. **A super node is created if**
A) a voltage source is common to two loops without a series resistance
B) a current source is common to two loops without a series resistance
C) a current source is common to two nodes without a series resistance
D) a voltage source is common to two nodes without a series resistance
6. **In a _____ circuit, the total resistance is smaller than the smallest resistance in the circuit.**
A) Series
B) Parallel
C) Neither series or parallel
D) None of the above
7. **The circuit is said to be in resonance if the current is _____ with the applied voltage.**
A) In Phase
B) 90° out of Phase
C) Out of Phase
D) 45° out of Phase
8. **Two wattmeter method of power measurement in 3 phase circuit is used in**
A) Balanced load only
B) Unbalanced load only
C) Both balanced and unbalanced
D) None of the above
9. _____ **allows the transient to dampen out and decay to the steady-state condition**
A) R
B) L
C) C
D) All of the above
10. **Inductor does not allow sudden change in**
A) Voltage
B) Frequency
C) Current
D) Phase

11. The unenergized capacitor at $t=0^-$ behaves as -----at $t=0^+$
 A) short circuit B) open circuit
 C) current source D) voltage source
12. The current in the R-C circuit at a time $t = 0^+$ is?
 A) V/R B) R/V
 C) V D) R
13. Laplace transform of an impulse is-----
 A) $1/s$ B) s
 C) 1 D) none of the above
14. Time constant of series Resistance and Capacitance circuit is-----
 A) R B) RC
 C) R/C D) C/R
15. If the roots of an equation are real and unequal, then the response will be?
 A) critically damped B) under damped
 C) over damped D) damped
16. The critical damping is said to occur when the frequency of damped vibration is
 A) Varying B) Increasing
 C) zero D) Decreasing
17. While considering Reciprocity theorem, we consider ratio of response to excitation as ratio of?
 A) voltage to voltage B) current to current
 C) voltage to current D) none of the mentioned
18. In superposition theorem when we consider one voltage source, all the other independent voltage sources are -----
 A) Shorted B) Removed
 C) Undisturbed D) Opened
19. If source impedance is complex, then maximum power transfer occurs when the load impedance is ----- the source impedance.
 A) complex conjugate of B) equal to
 C) negative of D) negative of complex conjugate of
20. A delta circuit has each element of value $R/2$. The equivalent elements of the star circuit will be
 A) $R/6$ B) $R/3$
 C) $2R$ D) $3R$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|------------|
| 1. a) Explain dot convention rule used in the coupled circuit. | 4 | L*2 | 2 | 1,2,5,9,12 |
| b) Explain the following terms.
(i) Quality factor
(ii) Half power frequencies
(iii) Bandwidth | 6 | L2 | 2 | 1,2,5,9,12 |
| c) Find V_1 and V_2 using nodal analysis for the circuit shown in figure 1c. | | | | |

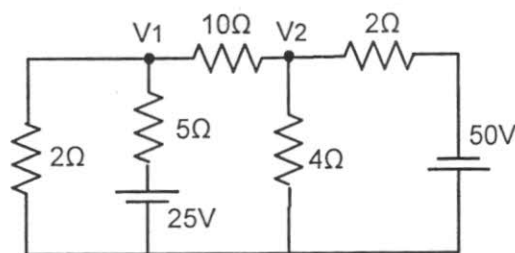


Figure 1c

6	L3	1	1,2,5,9,12
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2. a) Derive the equation for resonant frequency in series RLC circuit. Also plot various performance characteristics.
- b) A three phase 400V, 4-wire system has star connected load with RYB phase sequence. The three phase loads are given as $Z_R = 10\angle 0^\circ \Omega$, $Z_Y = (15 + j10)\Omega$ and $Z_B = 5\angle 90^\circ \Omega$. Find the line current and the current through the neutral conductor.
- c) Find the current through the capacitor using mesh analysis for the circuit shown in figure 2c.

4 L2 2 1,2,5,9,12

6 L3 2 1,2,5,9,12

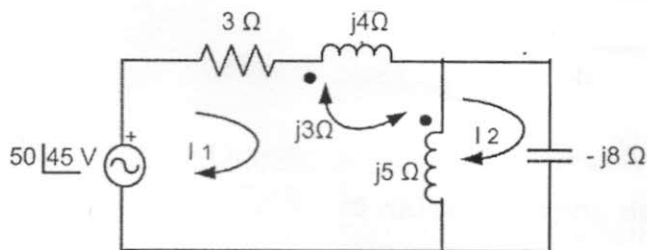


figure: 2c

6 L3 1 1,2,5,9,12

4 L2 1 1,2,5,9,12

6 L2 2 1,2,5,9,12

3. a) With an example, explain source transformation techniques
- b) Derive the expression for the neutral shift voltage in a three-phase unbalanced circuit.
- c) A coil of 20Ω resistance has an inductance of 0.2H and is connected in parallel with a condenser of $100\mu\text{F}$ capacitance. Calculate the frequency at which this circuit will be under resonance. Find also the value of dynamic impedance.

6 L3 2 1,2,5,9,12

Unit – II

4. a) Explain the behavior of inductor and capacitor at the instant of switching.
- b) A switch is moved from position 1 to position 2 in the circuit shown in figure 4b. The steady state is being obtained before switching. Calculate i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0^+$.

4 L3 3 1,2,5,9,12

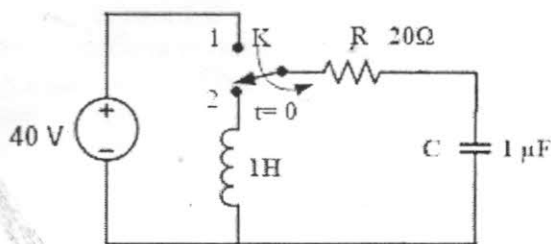


figure 4b

6 L3 3 1,2,5,9,12

6 L3 4 1,2,5,9,12

4 L1 4 1,2,5,9,12

- c) Derive the y parameters in terms of z parameters.
5. a) Define hybrid parameters.
- b) A network shown in figure 5b is under steady state with switch at the position 1. At $t=0$, switch is moved to position 2. Find $i(t)$.

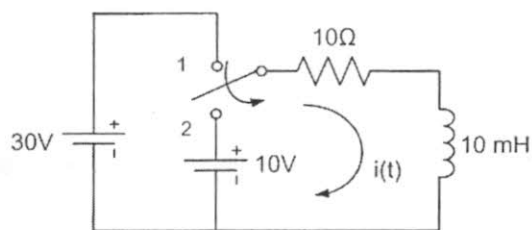


Figure 5b

6 L3 4 1,2,5,9,12

- c) Obtain the Laplace Transform of the following signal.

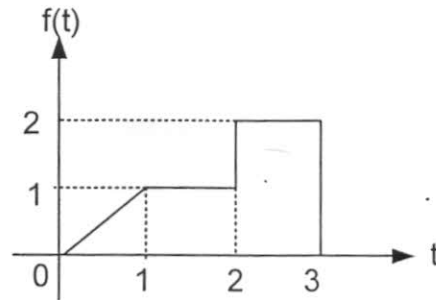


Figure 5c

6 L3 3 1,2,5,9,12

6. a) State and explain the following signals and hence obtain its Laplace transform.
(i) Step (ii) Ramp (iii) Impulse (iv) exponential
b) Find the ABCD parameters for the circuit shown in figure 6b.

8 L2 3 1,2,5,9,12

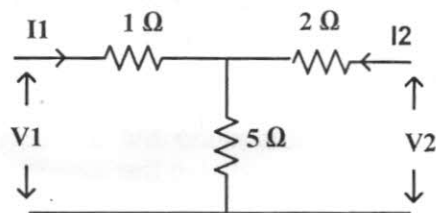


Figure:6b

8 L3 4 1,2,5,9,12

Unit – III

7. a) With an example, state and explain Thevenin's theorem.
b) Find the current through 5Ω resistor using reciprocity theorem for the circuit given in fig7b.

8 L2 5 1,2,5,9,12

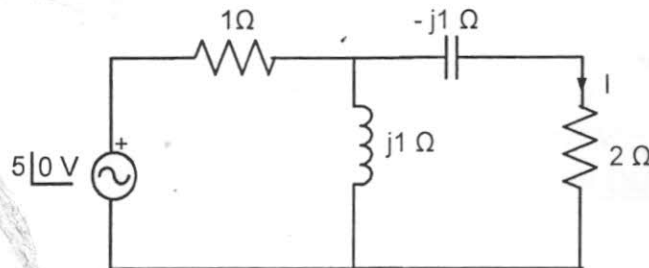


Fig: Q7b

8 L3 5 1,2,5,9,12

8. a) With an example state and explain Superposition theorem.
b) Determine the power supplied to the network shown in figure 8b.

8 L2 5 1,2,5,9,12

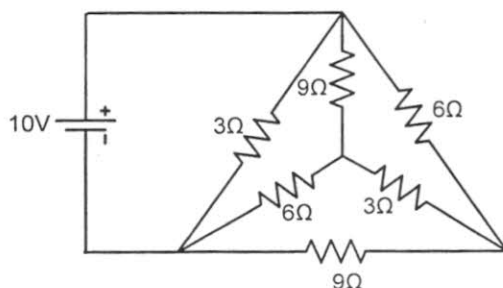


Figure:8b

8 L3 5 1,2,5,9,12

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
